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Quick Activity 2: Canteen Group Order

1. Identify the inputs, processes, and outputs.

Inputs	Processes	Outputs
Number of rice meals Number of drinks Payment amount	Calculate the rice-meal subtotal. Calculate the 10% discount on the rice-meal subtotal. Calculate the drink subtotal. Calculate the packaging fee. Subtract the discount from the rice-meal subtotal. Add the discounted rice meals, drinks, and packaging fee to get the final bill. Subtract the final bill from the payment to get the change.	Rice-meal subtotal Drink subtotal Discount Packaging fee Final bill Change

2. Decomposition

- Start.
- Enter the number of rice meals.
- Enter the number of drinks.
- Enter the payment amount.
- Calculate the rice-meal subtotal.
- Calculate the 10% discount.
- Calculate the drink subtotal.
- Calculate the packaging fee.
- Subtract the discount from the rice-meal subtotal.
- Add the discounted rice cost, drink cost, and packaging fee.
- Calculate the change.
- Display the results.
- End.

3. Write pseudocode or draw a flowchart.

START

READ riceMeals, drinks, payment

$\text{riceSubtotal} \leftarrow \text{riceMeals} \times 85$

$\text{discount} \leftarrow \text{riceSubtotal} \times 0.10$

$\text{drinkSubtotal} \leftarrow \text{drinks} \times 25$

$\text{packagingFee} \leftarrow \text{riceMeals} \times 5$

$\text{discountedRice} \leftarrow \text{riceSubtotal} - \text{discount}$

$\text{finalBill} \leftarrow \text{discountedRice} + \text{drinkSubtotal} + \text{packagingFee}$

$\text{change} \leftarrow \text{payment} - \text{finalBill}$

DISPLAY riceSubtotal

DISPLAY drinkSubtotal

DISPLAY discount

DISPLAY packagingFee

DISPLAY finalBill

DISPLAY change

END

4. Dry Run - Test Case 1

Given: 3 rice meals, 4 drinks, ₱500.00 payment

Step	Variable	Calculation	Result
1	riceSubtotal	$3 \times \text{₱}85.00$	₱255.00
2	drinkSubtotal	$4 \times \text{₱}25.00$	₱100.00
3	discount	$\text{₱}255.00 \times 0.10$	₱25.50
4	discountedRice	$\text{₱}255.00 - \text{₱}25.50$	₱229.50
5	packagingFee	$3 \times \text{₱}5.00$	₱15.00
6	finalBill	$\text{₱}229.50 + \text{₱}100.00 + \text{₱}15.00$	₱344.50
7	change	$\text{₱}500.00 - \text{₱}344.50$	₱155.50

Dry Run – Test Case 2

Given: 5 rice meals, 3 drinks, ₱600.00 payment

Step	Variable	Calculation	Result
1	riceSubtotal	$5 \times \text{₱}85.00$	₱425.00
2	drinkSubtotal	$3 \times \text{₱}25.00$	₱75.00
3	discount	$\text{₱}425.00 \times 0.10$	₱42.50
4	discountedRice	$\text{₱}425.00 - \text{₱}42.50$	₱382.50
5	packagingFee	$5 \times \text{₱}5.00$	₱25.00
6	finalBill	$\text{₱}382.50 + \text{₱}75.00 + \text{₱}25.00$	₱482.50

7	change	₱600.00 – ₱482.50	₱117.50
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Explain: Why would applying the 10% discount to the entire bill produce an incorrect result?

Applying the 10% discount to the entire bill would be incorrect because the discount only applies to the rice meals. The drinks and packaging fee do not have a discount. The 10% should only be calculated from the rice-meal subtotal.